

Master Theorem for Decreasing

Function
s

$$T(n) = T(n-1) + O(n)$$

$$T(n) = T(n-1) + n$$

$$O(n^2)$$

$$P(n) = T(n-1) + \log n$$

$$O(n \log n)$$

$$T(n) = 2T(n-1) + 1$$

$$= O(2^n)$$

$$T(n) = 3T(n-1) + 1$$

$$O(3^n)$$

$$T(n) = 2T(n-1) + n$$

$$=$$

Case *if*

General

Form

$$T(n) = a(T(n-b)) + f(n)$$

$$a > 0$$

$$a = 1$$

if $a > 1$

if $a < 1$

$$b > 0$$

and $f(n) =$
 (n^k)

$O(n^{k+1}) = O(n \cdot f(n))$.

where K_0

$O(n^k \cdot a^n)$ or $O(Cn^k$
 $a/b)$

$O(n) =$
 $(f(n))$

O

Master

Theorem

For Dividing
Function.

Gencial Form = $a \cdot T(A/b) +$

$f(n)$

$a > 1$

$O \log$
 $\log a$

O
 k

Case 1 :

$b > 1$

$f(n) = O(\log n)$

Case 2:

$\log a < k$
 $\log a$

$\log a = k$

k

b

then

0 (210969)

if $p > -1$

if $p = -1$

$C(n, t)$

login)

$\theta_{\text{СПК}}$

$\log$

$\log n;$

if $p < -1$

$O(n)$

Case 3: *if* $\log$,

ack

if $P \geq 0$ (nk
 $\log n$)
 C

if $P < 0$
 $O(nk)$

a = 2

$$T(n) = 4T(\sim) + 12$$

$$\log_2 24 = 2$$

$$T(n) = 2T(2) + 1$$

$$f(n) = O(1) = n^0$$

$$\log n$$

$$K = 0$$

$$\log_2 2$$

$$= 1$$

$$P=0$$

$$1K \quad C(n^2).$$

$$f(n) =$$

$$m' \log n$$

$$R=1 \quad P=0 \quad 22K$$

$$- O(n^2)$$

$$J(n) = 8 T(n)$$

$$) + n$$

$$\log_2$$

$$K = 1$$

$$K=!$$

$$8=3$$

$$P=0$$

$$\Theta(n^3)$$

$$T(n) = 9T$$

$$97 (1)$$

$$\log$$

$$39=2$$

$$T(a) = 87 (1) + n^2 8T(\frac{n}{2})$$

ame.

$$O(n^3)$$

+ 1

9 T (1
) + n

Same

C
(n2)

Case 2.

T
K=0 P20
27K
On2)

$\Pi (n) \quad \text{ЯТ} (13) +$

$$n^2 \log 2 = 1 = k =$$

$$\underline{1}$$

$$=== \underline{1}$$

$$P = 0 =)$$

$$O(n \log n)$$

$$T(n)$$

$$7(a) = 4T$$

$$(1) + m^2$$

2

2

$$\log y = 2 \approx k = 2$$

$$\text{Pro } (\log n)$$

$$T(n) = 4T$$

$$(\sim) + n^2$$

$$\log n \text{ (말)} n^2$$

login.

$$T(n) = 87 (2)$$

$$+ n^2$$

log

$$8=3$$

K23

Θ

$$O . n^3$$

$$\log n$$

Suppo
se

Case

3:

$\pi(n)$ 25MM

$2T(1)$

+

n

ГОД

И

$\log_2 2 = 1 =$

$K = 1$

$$P = -1$$

$$\Theta$$

$$O(n \log$$

$$n \log$$

$$\log n)$$

$$T(n) = 2 \left(T \left(\frac{n}{2} \right) \right) + T_{\text{gen}}$$

$$\log_2 2 = 1 \sim$$

$$K = 1$$

$$PL - 1$$

$$O(n)$$

$$T(n) = T(n)$$

$$+ n^2 \log_2 1 = 0$$

$$< K = 2 O(n^2)$$

$$J(n) = 2T$$

$$(1/2) +$$

Same

$\frac{n}{2}$
मर

$$F(x) = 25 (24)$$

$$+ n^2 \log_2 n$$

$$\log_7 = 12$$

$$K=2 \text{ dog?}=12$$

$$C(n^2$$

$$\log_2 n) \ominus$$

$$T(n) = 4T$$

$$(1) + n^3 \log_2$$

$$2 < K=3$$

$$O(\log_3)$$

(W= 45

(4)

magn

logn

0 (n3).